

Visualization, Intro

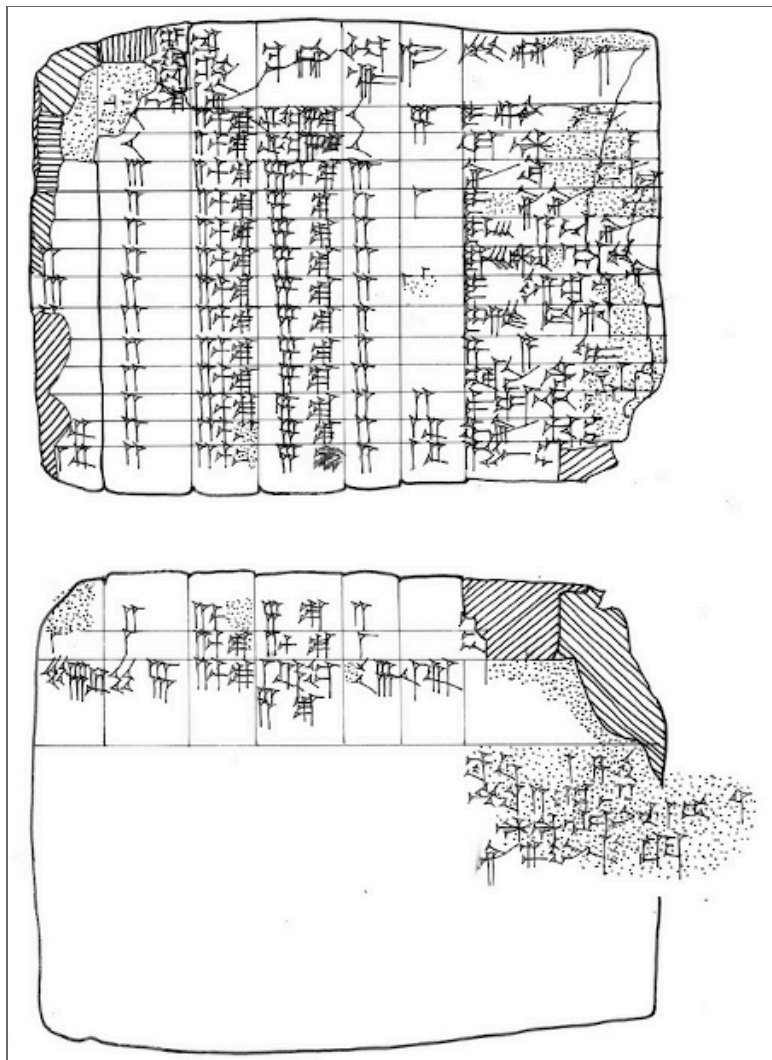
MGT 100 Week 1

Kenneth C. Wilbur and Daniel Yavorsky

Data Visualizations (Viz)

- “The greatest value of a picture is when it forces us to notice what we never expected to see.”
 - Boxplot Inventor John Tukey
 - Great visualizations raise new questions

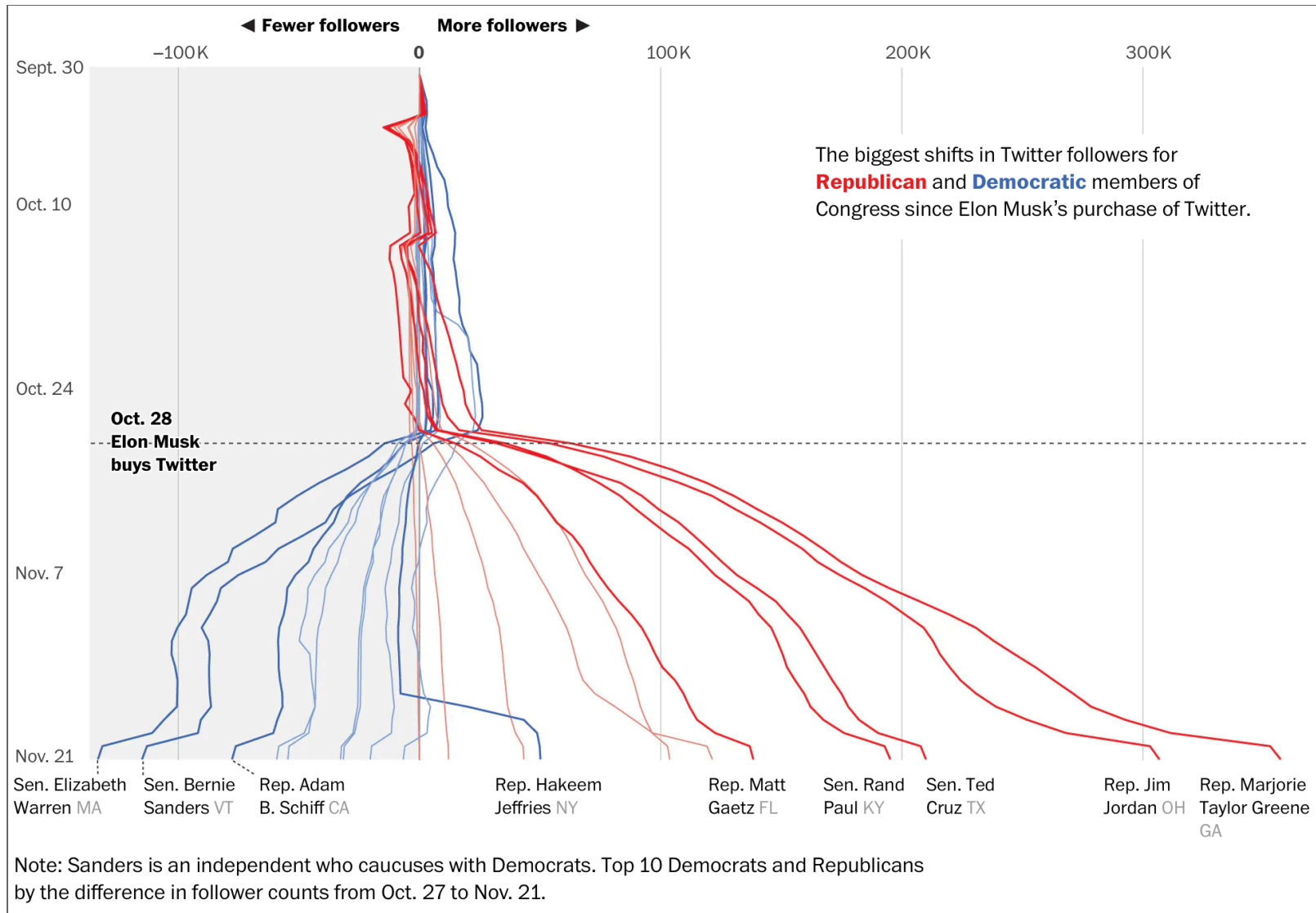
Prehistoric Data Viz



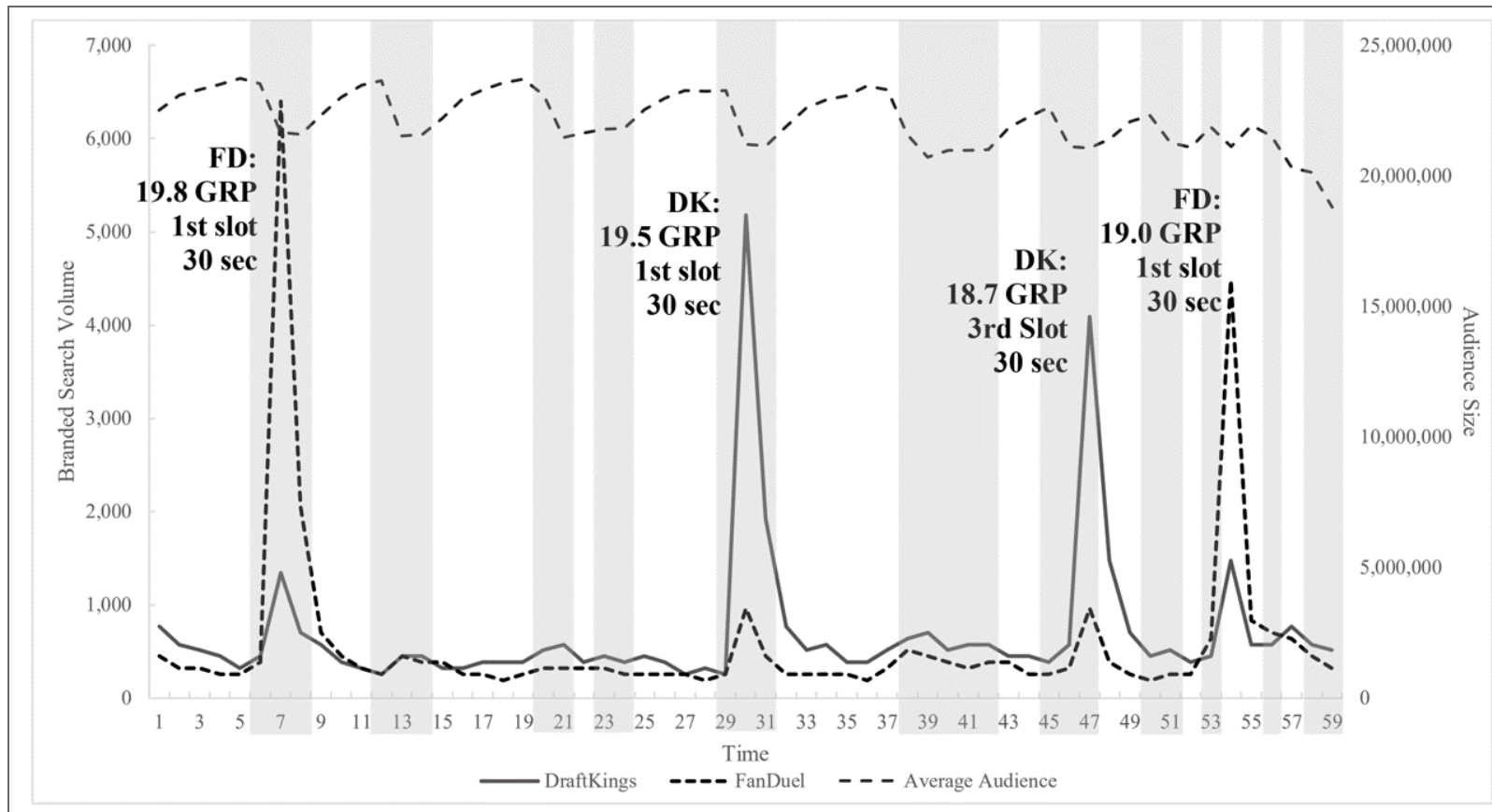
Translated

A	B	C	D	E	F	G
[15]	10	2½ s.	⅓ m. 5 s.	10	5	Igigi
[10]	10	2½ s.	⅓ m. 5 s.	10		Awat-Šamaš
[3]	3	2½ s.	7½ s.	3		Nidittum
[3]	2	2½ s.	5 s.	2	1	The sons of Ili-iddinam
[2]	2	2½ s.	5 s.	2		The sons of Ahuni
[2]	2	2½ s.	5 s.	2		The sons of Sin-šamuh
3	2	2½ s.	5 s.	2	1	The son of Imgurru
2	2	2½ s.	5 s.	2		Isi-dare
[2]	2	2½ s.	5 s.	2		Zayyalum
[2]	2	2½ s.	5 s.	2		Silli-Ištar
[4]	2	2½ s.	5 s.	2	2	Paqatiya
[3] ⅓	2	2½ s.	5 s.	2	1⅓	Perhum
[3] ⅓	2	2½ s.	5 s.	2	1⅓	Abum-ili
[2]	2	2½ s.	5 s.	2		[Sin-ašareda]
[1]	1	2½ s.	[2½ s.]	1		Aplum
57 [⅔]	46	2½ s.	1⅓ m. 5 s.	46	11⅓	
Tašritum (Month VI), [day...]						
Year: the third after Isin was destroyed by the great weapons of An, Enlil and Enki						
A = "Troops, available assets" B = "Those who were sent <i>from</i> Sin's dyke" C = "Earth, the (daily) work assignment of 1 man" D = "Its earth" E = "Check made" F = "Arrears" G = "Its name"						

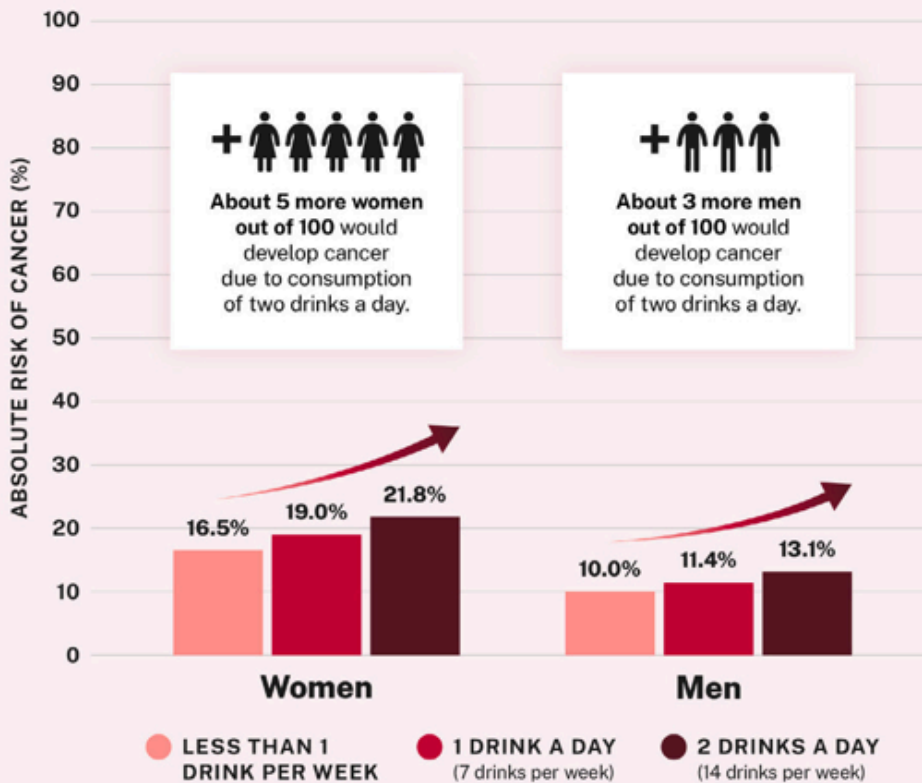
When Elon bought Twitter



DraftKings, FanDuel Searches during an NFL Game, 9-10 P.M.

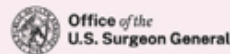


Higher alcohol consumption increases alcohol-related cancer risk in women and men

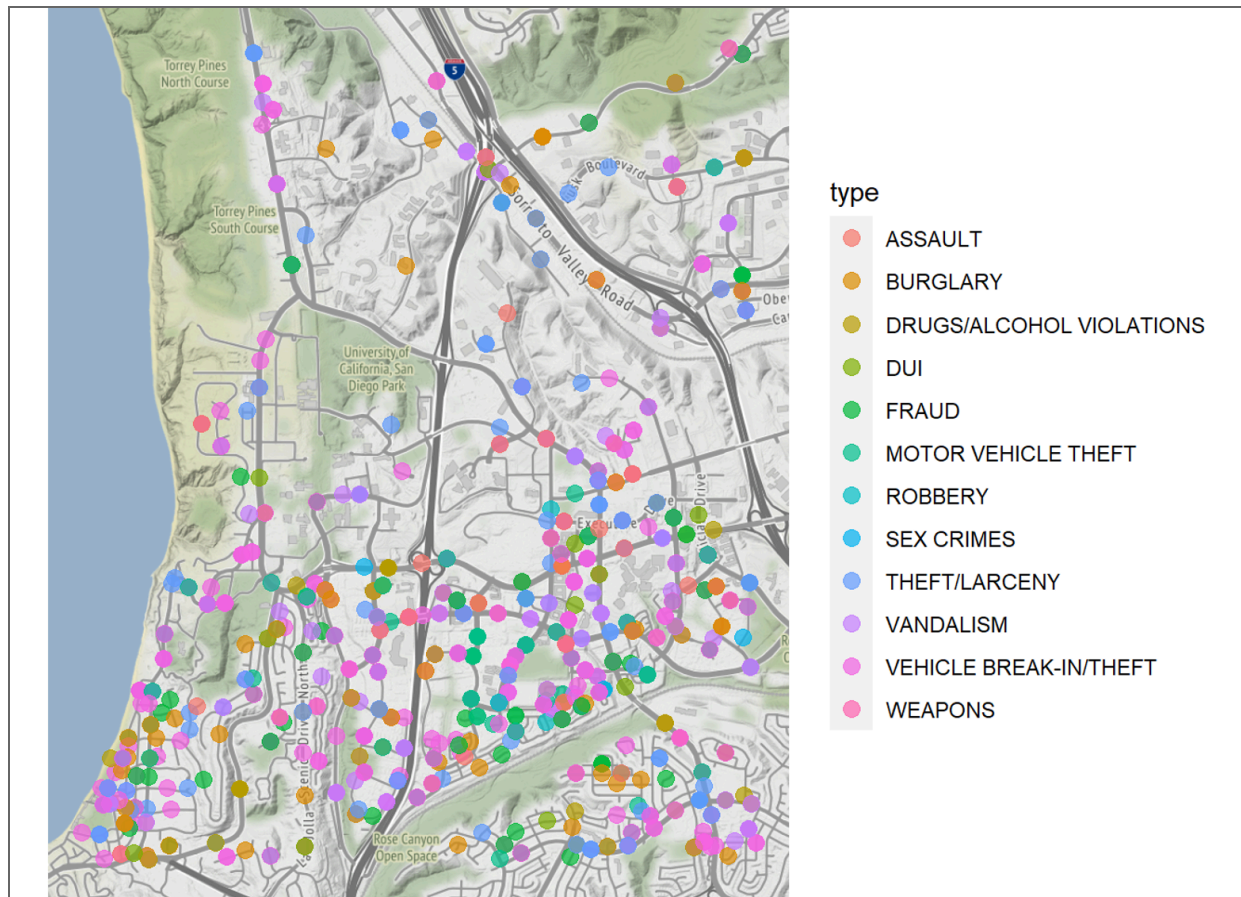


This graph represents the cumulative absolute risk of alcohol-related cancer in women and men over the lifespan by age 80. Alcohol-related cancer includes breast, colorectum, esophagus, liver, mouth, throat, and voice box cancers.

Source: Calculated with data from Sarich, P., Canfell, K., Egger, S., Banks, E., Joshy, G., Grogan, P., & Weber, M. F. (2021). Alcohol consumption, drinking patterns and cancer incidence in an Australian cohort of 226,162 participants aged 45 years and over. *British journal of cancer*, 124(2), 513–523. <https://doi.org/10.1038/s41416-020-01101-2>



SDPD Crime Reports Near UCSD



- When is data zero vs. missing? Provenance is key

Data *Provenance*

- describes people, entities, and activities involved in producing, compiling, transforming & sharing data
 - Verifiability requires source & cleaning scripts
- Crucial in determining quality, reliability, trustworthiness
 - Interpreting provenance requires and deepens domain expertise
- Important aspects: Customer consent; Data privacy; Missingness remedies; Contractually permissible uses
- When can you fully trust provenance information?
 - Best to treat provenance as hypotheses to be verified
 - Usually, provenance descriptions are missing or imperfect
 - Sometimes, provenance descriptions are marketing documents

Questions to ask

- Data Origin, Measures, Quality, Verifiability
 - How were the data produced?
 - Who collected it, when and why?
 - How is each variable measured? (surveys, APIs, sensors, etc; units)
 - Is there a data dictionary or changelog?
 - How can I verify individual datapoints?
 - Is the data certified by any third party?
- Processing & Transformation
 - What cleaning or manipulation was done?
 - Who performed it and why? Can I see their scripts?
 - Were missing values imputed? How?
 - How were outliers handled?
- GIGO: Garbage In, Garbage Out. Don't analyze noise

Viz can build trust, understanding

- Eyeballs can interpret pictures quickly
 - We all interpret visual patterns and make predictions
- Can detect unknown errors
- Understandable to managers
 - Viz choices enable narrative understanding
- Viz are *lingua franca* across disciplines
 - Easily replicated -\> more easily trusted
- Successful viz raise deeper questions
 - Asking the next question indicates acceptance
 - "I wonder why that is...." or "Maybe that's because..."

Number of World Heritage Sites

Norway

Denmark

Sweden

2004

5

4

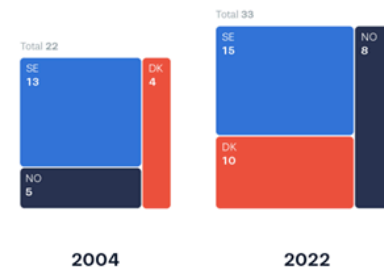
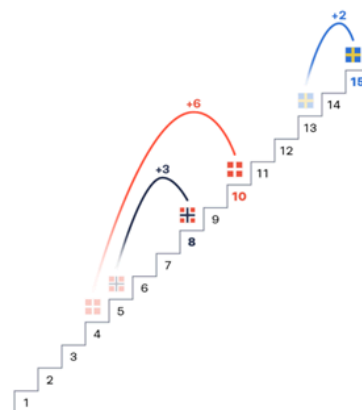
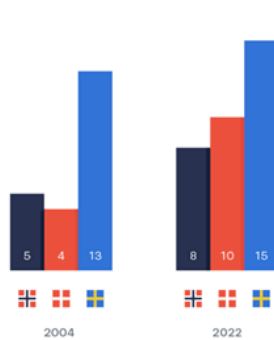
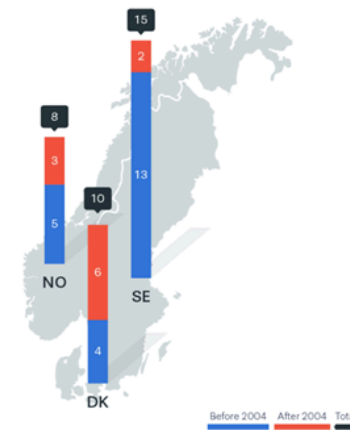
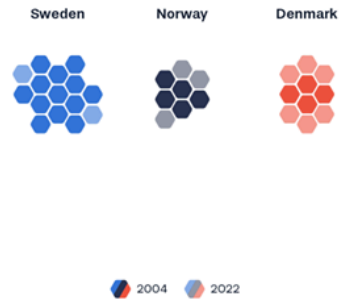
13

2022

8

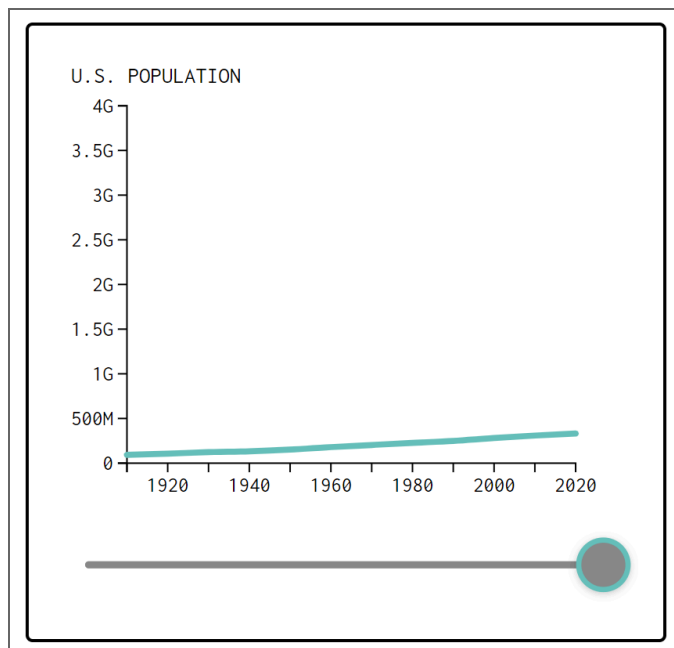
10

15



Honest Viz

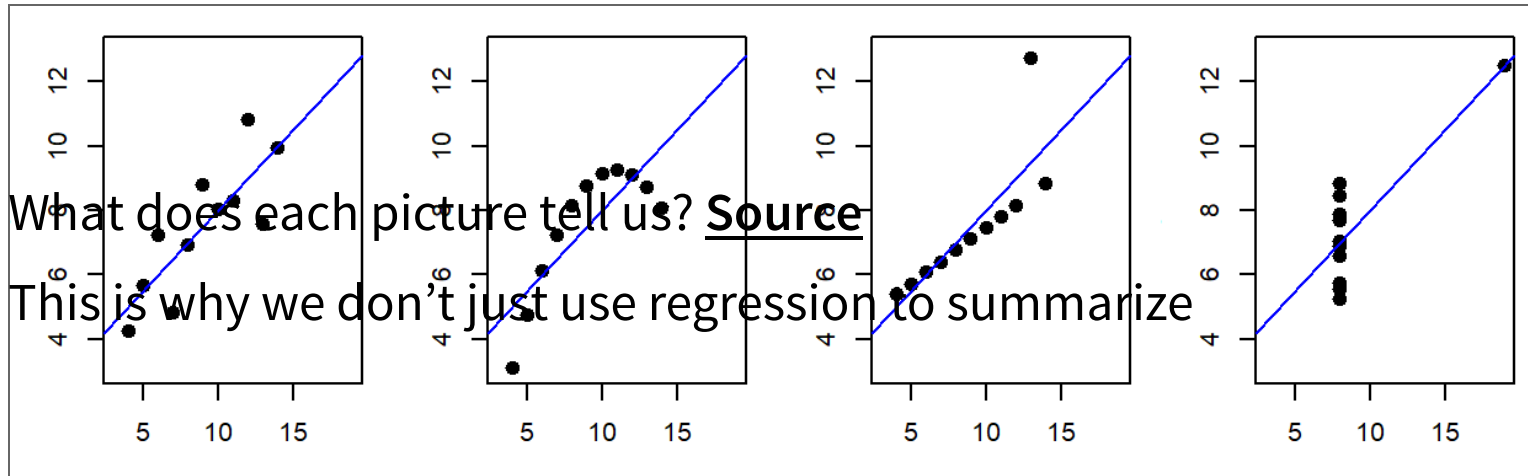
1. Start axes from 0, choose scales judiciously
2. Show all relevant data, label accurately. Don't edit, trim
3. Discretize and smoothe judiciously



Visualize data before you analyze

- 4 Datasets, Same $\hat{\beta}^{OLS}$, i.e. same $\frac{x'x}{x'y}$

- What does each picture tell us? Source
- This is why we don't just use regression to summarize



Customer Analytics

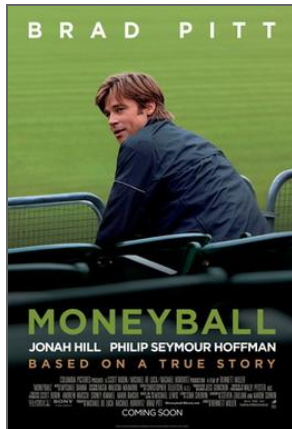
Customer

- Receives good or service in exchange for payment (money, time, attention)
- Has agency: Can say “no”
- “The purpose of business is to create and keep a customer.”
-Drucker



Analytics

- Using data to improve decisions
- Started by Charles Taylor in the 1880s
- Popularized by *Moneyball* (2011)
- Measurement, Heuristics, Graphics, Models, Predictions, Automation, ...
- Can be deceptively difficult



First Law of Customer Analytics

- No Customers, No Business
- No Customers ->
No Revenue ->
No Profit ->
No Business

Second Law of Customer Analytics

- Mo' Customers, Mo' Profits
- More Customers ->
More Revenue ->
More Profit
 - These are empirical tendencies, not logical necessities
 - Individual customers can be unprofitable if $(\text{price} - \text{cost}) < 0$
- Objective: Maximize Long-term Profits
 - Long-term focus aligns our interest with customers
 - Sidesteps or reduces most ethical dilemmas

Example of Great Marketing: Netflix

- Consumer surplus

- Suppose customer pays \$20/month, watches 60 hours: \$0.33 per entertainment hour, or \$0.13/hour with ads, or less with shared accounts
- A la carte rentals are more like \$1+/hour
- Non-video entertainment is much more expensive
- Social benefits may accrue from shared viewing
- Other streaming services may be competitive

- Producer surplus

- NFLX spent \$13B on content in 2023, 230 million subscribers, about \$4.70/user/month
- Cost structure: High fixed, low marginal
- Ads likely earning around \$4-5 per recipient/month
- Net profit margin about 20-22% in a competitive category

How can we use customer data?

- Businesses have only four ways to make money:
Acquire, develop, retain and “fire” customers

This is called Customer relationship management (CRM)

- Marketing mix: Improve product offerings, prices, promotion, distribution
- Understand customer heterogeneity for targeting, personalization, recommendations, product development...
- Privacy and security, e.g. misuse, theft, regulatory compliance

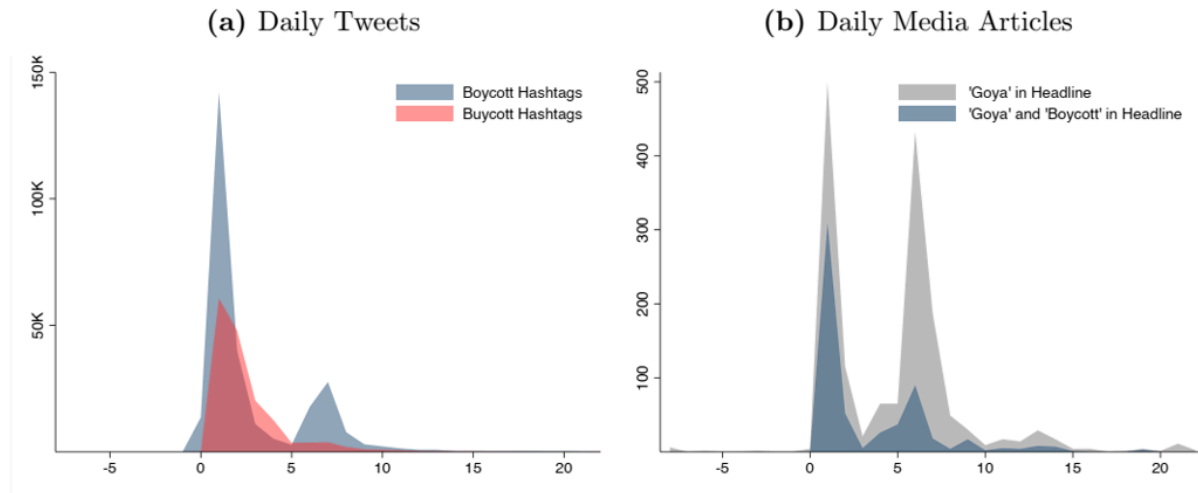
Example: Nielsen

CONSUMER PANEL DATA

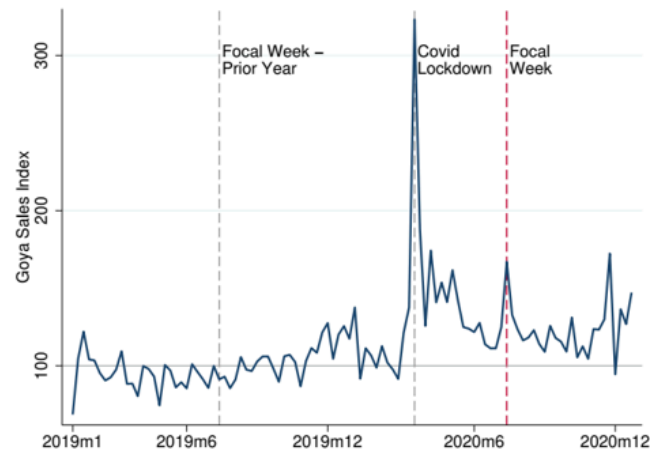
- The Consumer Panel Data include longitudinal data beginning in 2004 with annual updates. These data track a panel of 40,000–60,000 US households and their purchases of fast-moving consumer goods from a wide range of retail outlets across all US markets.

RETAIL SCANNER DATA

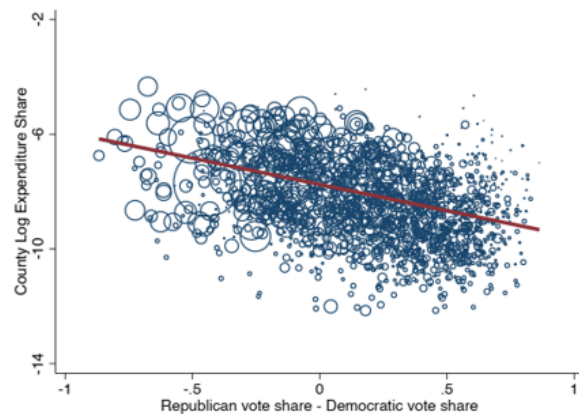
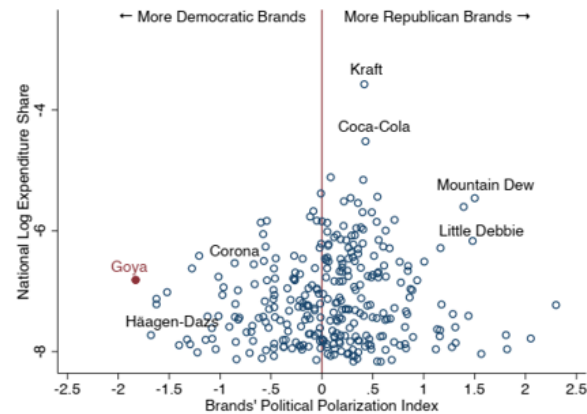
- Retail Scanner Data consist of weekly pricing, volume, and store environment information generated by point-of-sale systems from more than 90 participating retail chains across all US markets. Data begin in 2006 and include annual updates.

Figure 1: Images of Tweets Triggering Goya Boycott and Buycott**Figure 2: Goya Related Daily Tweets and Media Coverage in July 2020**

Notes: Day 0 is July 9th, 2020. Panel (a) depicts the number of daily tweets in July 2020 that use boycott- or buycott-related hashtags. Web Appendix A.3 details the classification procedure and Table A1 lists the hashtags. Panel (b) plots the number of daily media articles that include 'Goya' and 'Boycott' in the headline. Only 18 articles included buycott related keywords so they are not depicted. In both panels, the second spike is triggered by President Trump and his daughter's tweets in support of Goya.

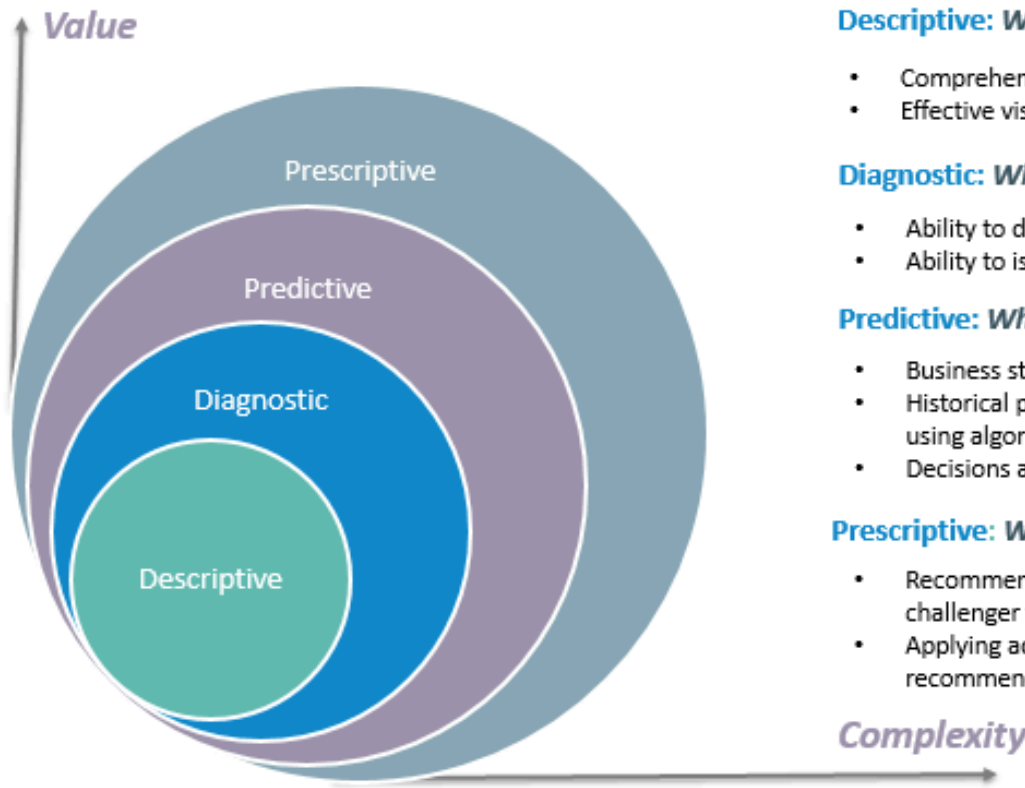
Figure 3: Total Weekly Spending Index on Goya Products

Notes: This figure depicts weekly spending index on Goya products from 2019-2020. The pre-pandemic sales average is normalized to 100. The red vertical dashed line indicates the focal week of the scandal (July 9-15). The spike and subsequent dip in November 2020 is due to bunching in purchases around the Thanksgiving holiday. Please see Appendix B.8 for a replication of the sales trend using Nielsen RMS data.

Figure 5: Polarization in Brand Popularity: Goya vs Other Top Brands**(a) Goya Market Share and Vote Margin****(b) Polarization of Goya and Other Top Brands**

Notes: Panel (a) depicts the relationship between Goya's log expenditure share in a given county in 2019 and the vote margin in that county. The size of the bubble is proportional to the number of panelists in a given county. Panel (b) plots the estimated political polarization index for each of the top 301 CPG brands. The index quantifies the extent to which the market share of a brand varies with political preferences. Goya's national expenditure share is 0.11% (the log of its national expenditure share is -6.81). See Web Appendix A.1 for more detail.

4 types of Data Analytics



What is the data telling you?

Descriptive: *What's happening in my business?*

- Comprehensive, accurate and live data
- Effective visualisation

Diagnostic: *Why is it happening?*

- Ability to drill down to the root-cause
- Ability to isolate all confounding information

Predictive: *What's likely to happen?*

- Business strategies have remained fairly consistent over time
- Historical patterns being used to predict specific outcomes using algorithms
- Decisions are automated using algorithms and technology

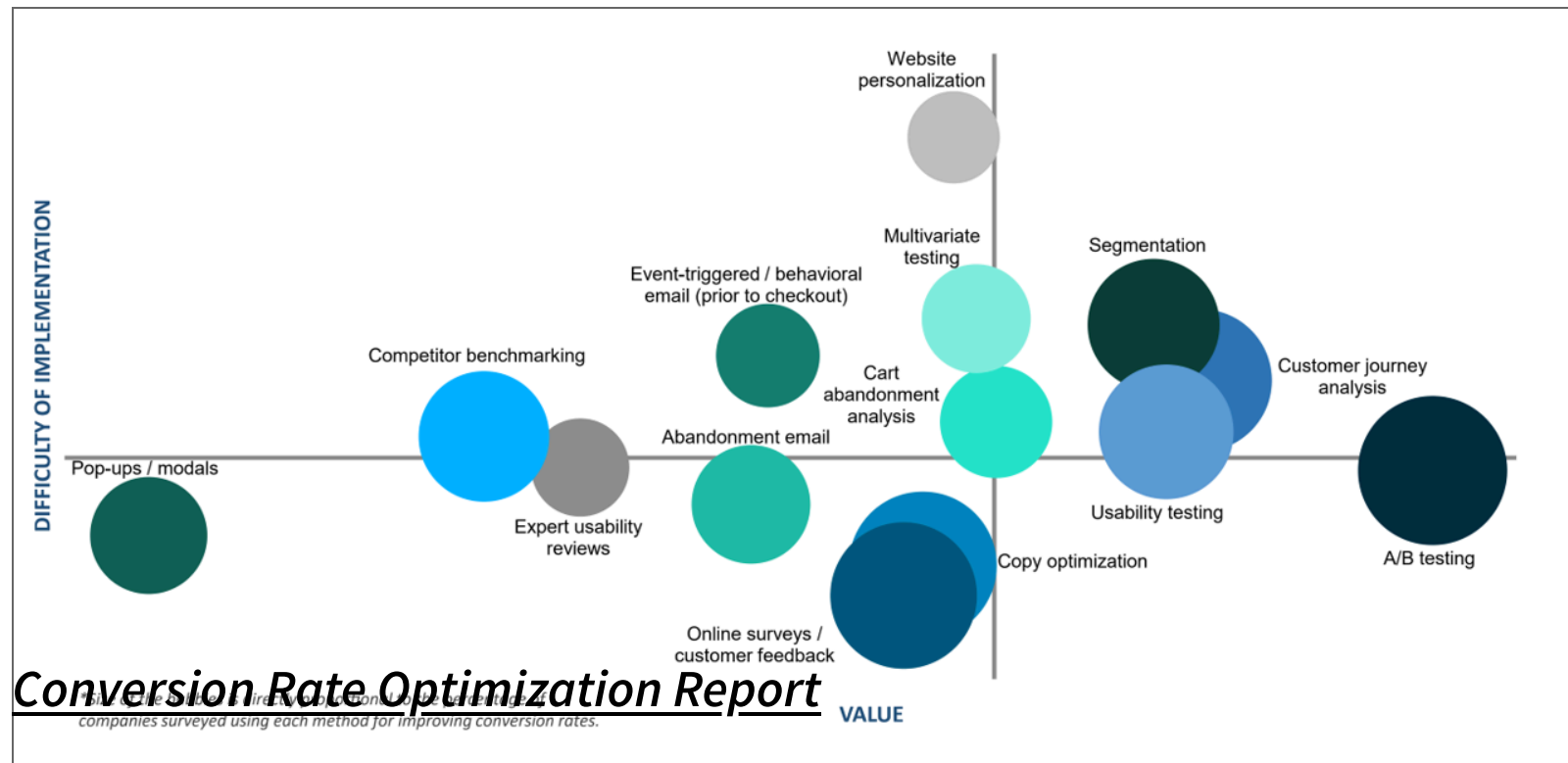
Prescriptive: *What do I need to do?*

- Recommended actions and strategies based on champion / challenger testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

E-commerce Funnel



E-commerce Analytics



ers?

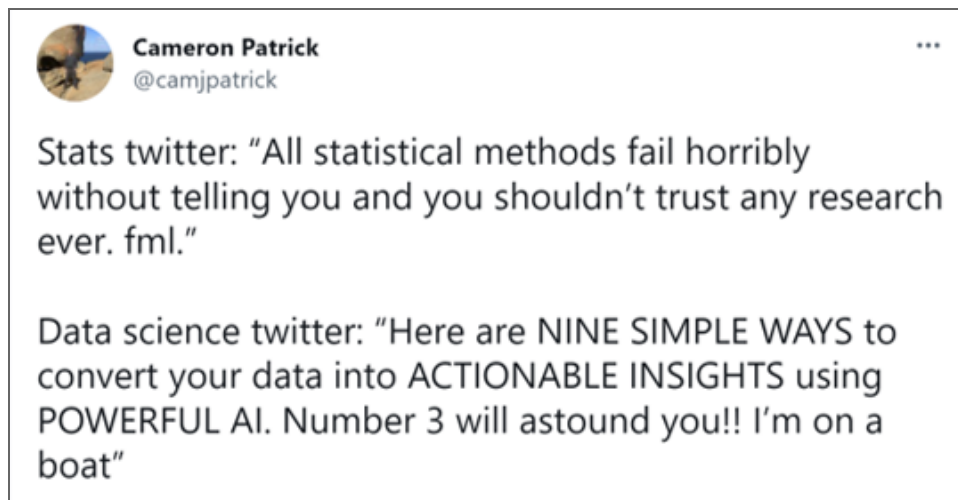
Using Customer Data for Customer Analytics

How do we “do” customer analytics?

- Decide objectives & outcome metrics
- Collect, wrangle, clean & verify relevant data
- Analyze data
- Communicate analyses and recommendations
- Make decisions
- Implement data-driven decisions
- Retrospectively evaluate and improve
- Repeat
- ...Once you have a stable process, automate carefully & monitor

Challenges: Executives

- May be territorial, or incentivized to be
- May worry that analytics will constrain or replace them
- May think data == magic
- May prefer hunches or misunderstand uncertainty



Challenges: Analysts

- Expensive
- Hard to find
- Not always current
- Not always interested in business

Challenges: Cultural

- Do analytics make or justify decisions?
- High- or low-trust environment? Tolerance for uncertainty?
 - You have limited credibility. You may only get a few strikes
- Do messengers get rewarded or shot?
- Are data available and integrated?
- Do teams work together or compete?

Signs of a great analytics org

- C-level champion(s), i.e. C{D,E,A,F,M,O}O
- Centralized team regulates data, arch., standards & tools
- Decentralized analysts collaborate with execs
- Analytics career tracks are well established
- Careful in-housing/outourcing decisions about analytics
- Good examples?

Analytics truisms

- Analytics matters more in B2C than B2B (why?)
- Selection effects are usually large

treatment effects are usually small

- Demographics don't predict behavior very well
- Agencies lie about data sometimes
- “If it's written in LaTeX, it's probably correct”

MGT 100

Course Design Principles

- Survey the field with pointers for deeper learning
- Focus on understanding:
“When it comes to LLMs, skillful prompting leaves amateurs in the dust.”
- Strict communication policies
 1. Website for class content
 2. Canvas for study groups & grades
 3. Piazza for all asynchronous interaction. No email or canvas messages
 4. After class, break or office hours for live discussions
- Course site & schedule

Gen AI effects on productivity

Pulling up the ladder

Impact of generative AI on the gap between high- and low-performing workers

Study	Topic	Inequality
Peng et al. (2023)	Coding efficiency	↓
Brynjolfsson, Li and Raymond (2023)	Customer chat	↓
Noy and Zhang (2023)	Writing quality	↓
Dell'Acqua et al. (2023)	Product design	↓
Chen and Chan (2023)	Ad effectiveness	↓
Choi, Monahan and Schwarcz (2023)	Legal analysis	↓
Otis et al. (2023)	Profits and revenue	↑
Roldan-Mones (2024)	Debating points	↑
Toner-Rodgers (2024)	Material discovery	↑
Kim et al. (2024)	Investment decisions	↑

Source: *The Economist*

Tips to max your learning

- Read the syllabus carefully
- Contribute as suggested in the syllabus
- Budget 5-10 hours/week
- Between classes:
 1. Understand script, do homework
 2. Check homework with group, resolve differences
 3. Monitor Piazza and read for the next class

Studying and Integrity

- We assign attending students to study groups in week 2
- It is OK to share homework scripts and answers
 - We do not collect or assess homeworks
 - Exam questions will test your familiarity and understanding of homework answers
- We encourage you to use Gen AI; we use it too
 - Be advised, you may get what you pay for
 - Free models are sometimes worse than useless



- Please write down your intentions for this class.
- How will you measure your effort?
- How will you know you are doing it right? Why is it required, plan ahead.

Vocab

Common language helps communication

Customer level

- Core need: identifiable problem a customer wants to solve. Could be functional, emotional, social, profit-motivated, etc. Related: desire, want, pain point
- Core benefit: Customer's desired outcome of a purchase. E.g., commuters need to get to school, not necessarily cars
- Consumer: Entity that experiences the core benefit
- Customer: Entity that purchases and pays

Product level

- Product/service/experience: Distinct offering that provides the core benefit
- Features: Aspects of a product that provide additional tangible or intangible benefits
- Contribution margin: $\text{Price} - \text{marginal cost}$
- Competitor: Any paid or free alternative that addresses the core need. E.g., commute by bike, walk, bus, trolley, Uber, scooter, skateboard; work from home

Market level

- Market: A group of potential customers with the same core need
- Segment: Distinct subgroup of similar customers
- Targeting: Which segment(s) a firm tries to serve
- Positioning: Specification of product features to suit targeted segments
- Marketing: Practice of meeting customer needs profitably **How to be good at marketing**

Legacy Terms

- 3/4/5 C's: Customer, Competitor, Company; Context; Complementors
- 4/.../10 P's: Price, Product, Promotion, Place AKA distribution

Client

Coding & Script

Coding errors

- “The good news about computers is that they do what you tell them to do. The bad news is that they do what you tell them to do.”
- Conjecture:
(debugging difficulty) is exponential in (lines of code)
- We can code fast or slow

Coding Habits

- Good habit: Test chunks as you code
- Test = Check that output matches expectation
- “Go slow to go fast”

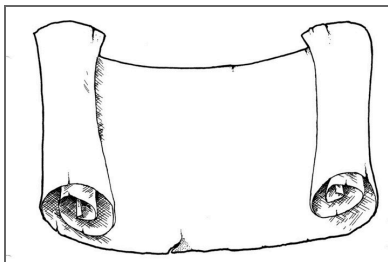


Pipe |>

- $y \leftarrow f(g(x))$ is the same as
- $y \leftarrow x \mid >$
 $g \mid >$
 f
- Why?
- Old pipe was `%>%`

Today's script

- Install/update R and Rstudio
- Posit.Cloud
- Data Import/export
- Data manipulation, summarization
- 5 verbs: Summarize, select, filter, arrange, mutate, group_by
- Univariate statistics
- Univariate plots
- Bivariate statistics
- Bivariate plots



- the script, run it, see where it breaks
- Digression: Production code vs prototype code

Wrapping up

Recap

- Data Viz should be the first step in customer analytics
- Customer analytics :
Using customer data to improve decisions
- Marketing : Meeting customer needs profitably
- Analytics types:
Descriptive, Diagnostic, Predictive, Prescriptive
- Summarize, select, filter, arrange, mutate, group_by



Going further

- Marketing Analytics for Data-Rich Environments
- GGplot2 YT video
- *R for Data Science (2e)*
- *R for Marketing Research and Analytics*
- GGplot2 Book
- *Advanced R*

